

DVCON

Stage 2A — Approach Description

K Jyothisna Padma | B Mythri Reddy | MSN Subhiksha | SpecOps 199, NIT Warangal

1. Overview

The Stage 1 proposal introduced a task-aware object detection framework built on a heterogeneous CPU–FPGA platform. The core thesis was that conventional detectors are task-agnostic — they enumerate every visible object with equal priority — whereas real applications require only the subset of objects most relevant to a given intent. Stage 2A realizes this proposal in software, preserving the exact algorithmic structure while substituting an accessible compute substrate for the target hardware.

2. Proposal vs. Implementation

The Stage 1 design partitioned the pipeline into four well-defined stages (see Section IV of the proposal). Each stage maps directly to an implemented component:

Stage	Component	Role in Pipeline
A — Detection	YOLOv8n (Detection Head)	Bounding boxes + class labels + confidence scores
B — Feature Extraction	EfficientNet-B0 (CNN Backbone)	80-dim feature vector x_i per detected crop
C — Task Scoring	Dot-Product Scorer (VEGA CPU)	$\text{score}_i = x_i \cdot W_{\square} \rightarrow \text{ranked object list}$
D — Selection	Relative Thresholding	Keep objects with $\text{score} \geq 0.70 \times \text{max_score (top-k)}$

The proposal specified EfficientNet as the CNN backbone — chosen for its compound scaling of depth, width, and resolution and its suitability for post-quantization FPGA deployment. This is honoured in Stage 2A: EfficientNet-B0 (torchvision) is loaded as a feature extractor, its classification head removed, and its 1280-dim output embedding projected to an 80-dim COCO-aligned feature vector x_i via a reshape–L2-norm operation. YOLOv8n supplies the bounding boxes, playing the role of the ‘detection head’ block shown in the Stage 1 block diagram.

3. EfficientNet-B0 Feature Vector Construction

For each bounding box detected by YOLOv8n, the pipeline:

- **Crops** the detected region from the input image.
- **Resizes** the crop to 224×224 and normalises with ImageNet mean/std.
- **Passes** it through EfficientNet-B0 to obtain a 1280-dim embedding.
- **Reshapes** the embedding to (80, 16) and computes the L2 norm along the 16-dim axis, yielding an 80-dim projection aligned with the 80 COCO class slots.
- **Reinforces** the detected class channel: $x_i[c] = \max(\text{projection}[c], \text{confidence} + \text{area_bonus})$, ensuring the primary class signal dominates while EfficientNet context contributes to adjacent dimensions.

This construction faithfully implements the feature vector x_i described in the proposal’s Relevance Scoring section, where x_i encodes both object identity and spatial prominence.

4. Task-Aware Scoring and Selection

Task weight vectors $W_{\square}$ ($t = 1 \dots 14$) are defined in `task_config.py`, exactly as specified in the Stage 1 proposal. Each 80-dim vector assigns a relevance weight to every COCO class for the given task. The relevance score is computed as:

$$\text{score}_i = 0.5 \times (x_i \cdot W_{\square}) + 0.5 \times (W_{\square}[c] \times \text{confidence}_i)$$

Relative thresholding then retains all detections whose score exceeds 70% of the maximum score, up to top-5 candidates. The highest-scoring object is presented as the primary answer (gold bounding box); others appear as valid alternatives (green). This matches the proposal's Stage IV description of relative thresholding as a mechanism that "avoids the limitations of fixed thresholds and ensures that multiple relevant detections are retained."

5. Demonstration Result

Running the pipeline on a kitchen image for task "water plant" (Task 5, W_5) produced the following ranked output:

- **BEST — bowl** (score = 0.6455, $W_5[\text{bowl}] = 0.50$)
- **VALID — cup** (score = 0.5898, $W_5[\text{cup}] = 0.70$)
- **VALID — vase** (score = 0.4924, $W_5[\text{vase}] = 0.60$)

The annotated output image shows gold/green bounding boxes on task-relevant objects and gray boxes on irrelevant ones (oven, chair), confirming that the pipeline correctly prioritises objects by task context rather than raw detection confidence.







